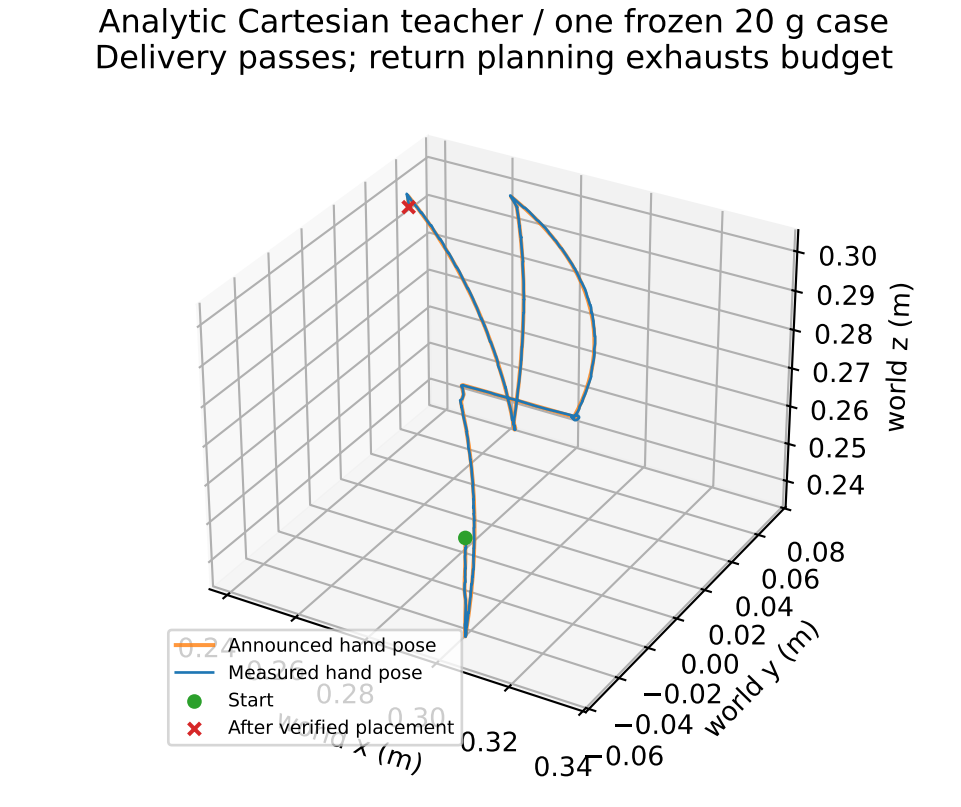
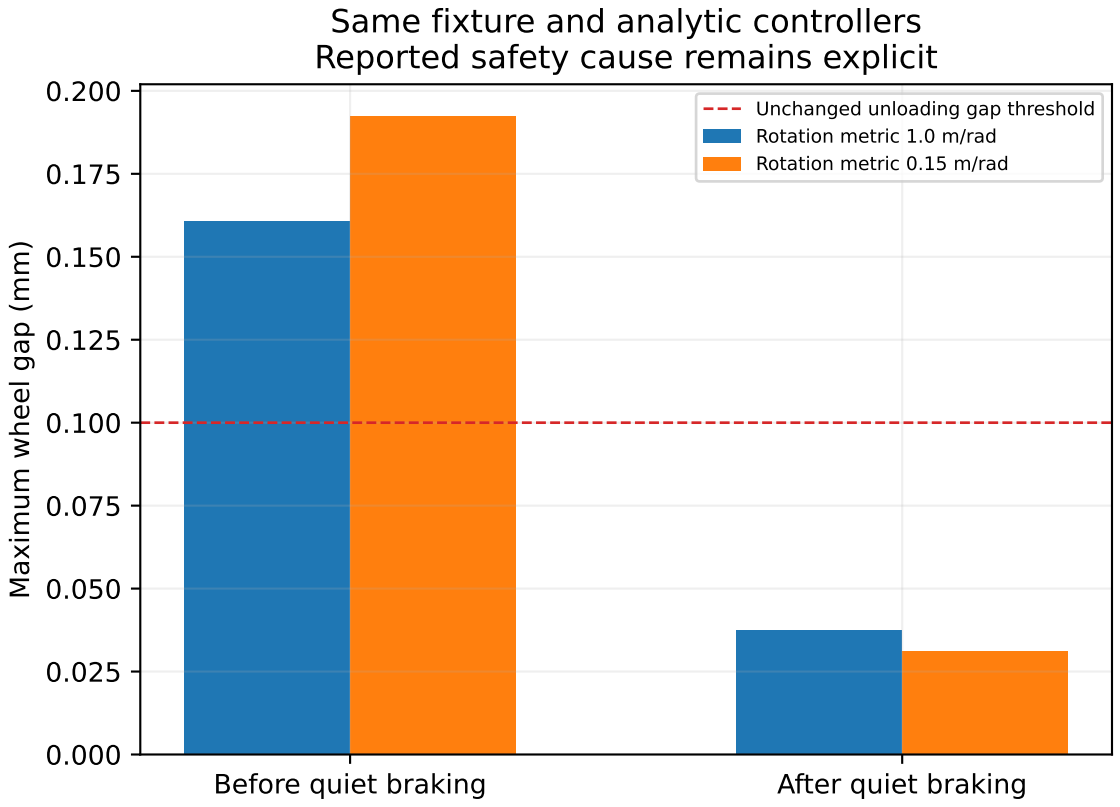

Analytic Cartesian teacher / one frozen 20 g case
Delivery passes; return planning exhausts budget
Announced hand pose
Measured hand pose
Start
After verified placement
world z (m)
world y (m)
0.30
0.29
0.28
0.27
0.26
0.25
0.24
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
0.32
0.34


Same fixture and analytic controllers
Reported safety cause remains explicit
Unchanged unloading gap threshold
Rotation metric 1.0 m/rad
Rotation metric 0.15 m/rad
Maximum wheel gap (mm)
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Before quiet braking
After quiet braking